



Performance Curves and Surge Control

DRESSER-RAND

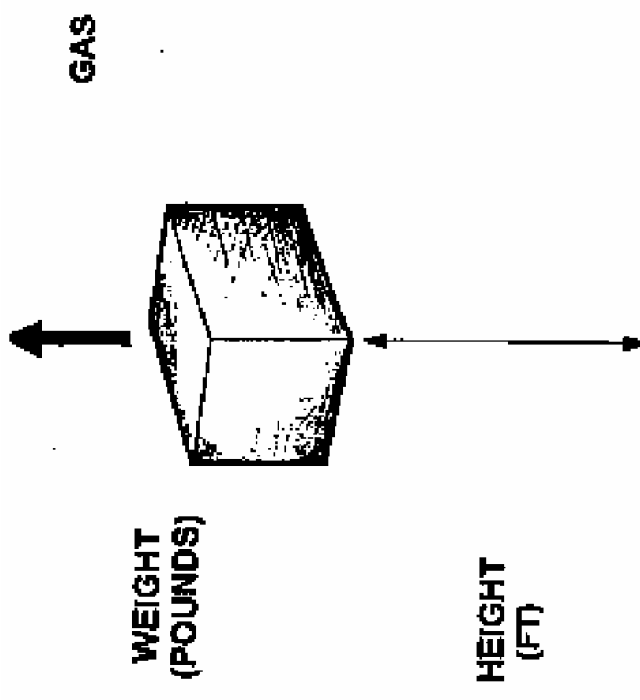
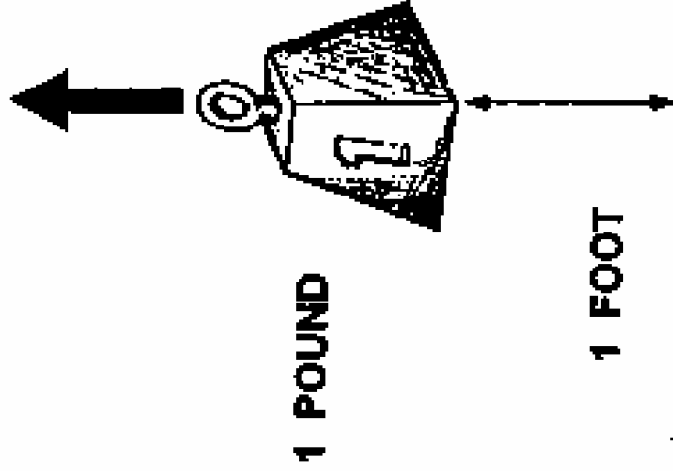
Performance Curves

- ◆ Head Concept
- ◆ Basic Components
- ◆ Fixed/Variable Speed
- ◆ Surge/Overload
- ◆ Effects on Performance



Head Concept

- ◆ **Mechanical:** The “work” (energy) developed to raise a weight of 1 pound by a distance of one (1) foot. Expressed in *foot-pound* (or equivalent Kgm or Nm);
- ◆ **Gas Compressors:** “work” done by the compressor / amount of gas. The head expressed in feet, is the height to which the gas could be lifted



Head Concept

- ◆ The height to which the gas is lifted *depends* on the *velocity* of the gas
- ◆ For any given RPM, the head developed by the compressor is fairly *constant, independent of the gas nature*.
- ◆ Head is depending upon:
 - Compressor geometry (i.e. no of stages, impeller diameters)
 - Compressor speed

$$H = ZRT_s \frac{M}{M-1}$$

$$M: \frac{K-1}{K \eta}$$

$$\text{where: } K = \text{Isentropic Exponent}$$

$$\eta = \text{Efficiency}$$

Z: Compressibility Factor

R: Gas Constant = 1545 / MW

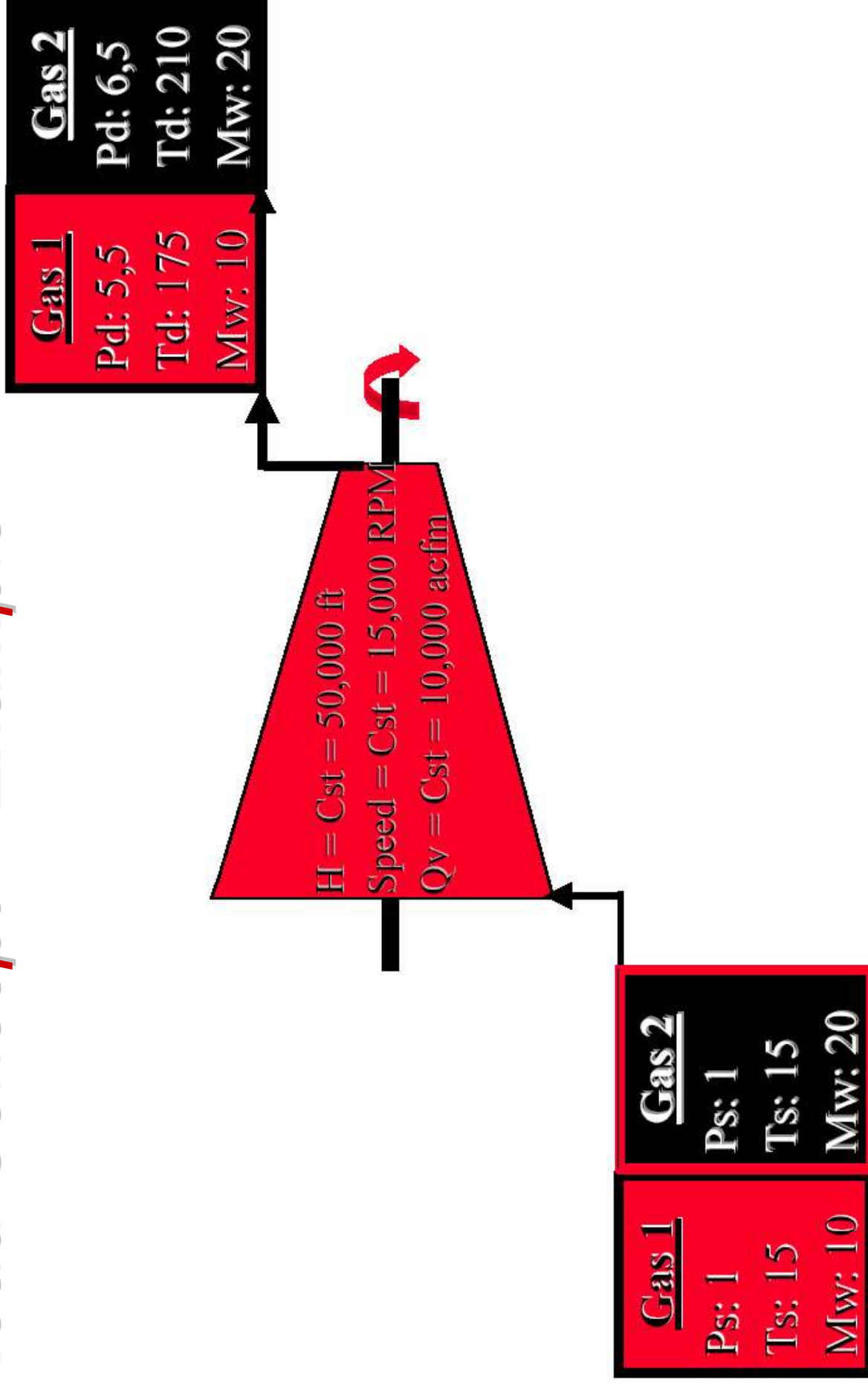
T_s: Suction Temperature (°R)

r: Pressure Ratio (P_d / P_s)

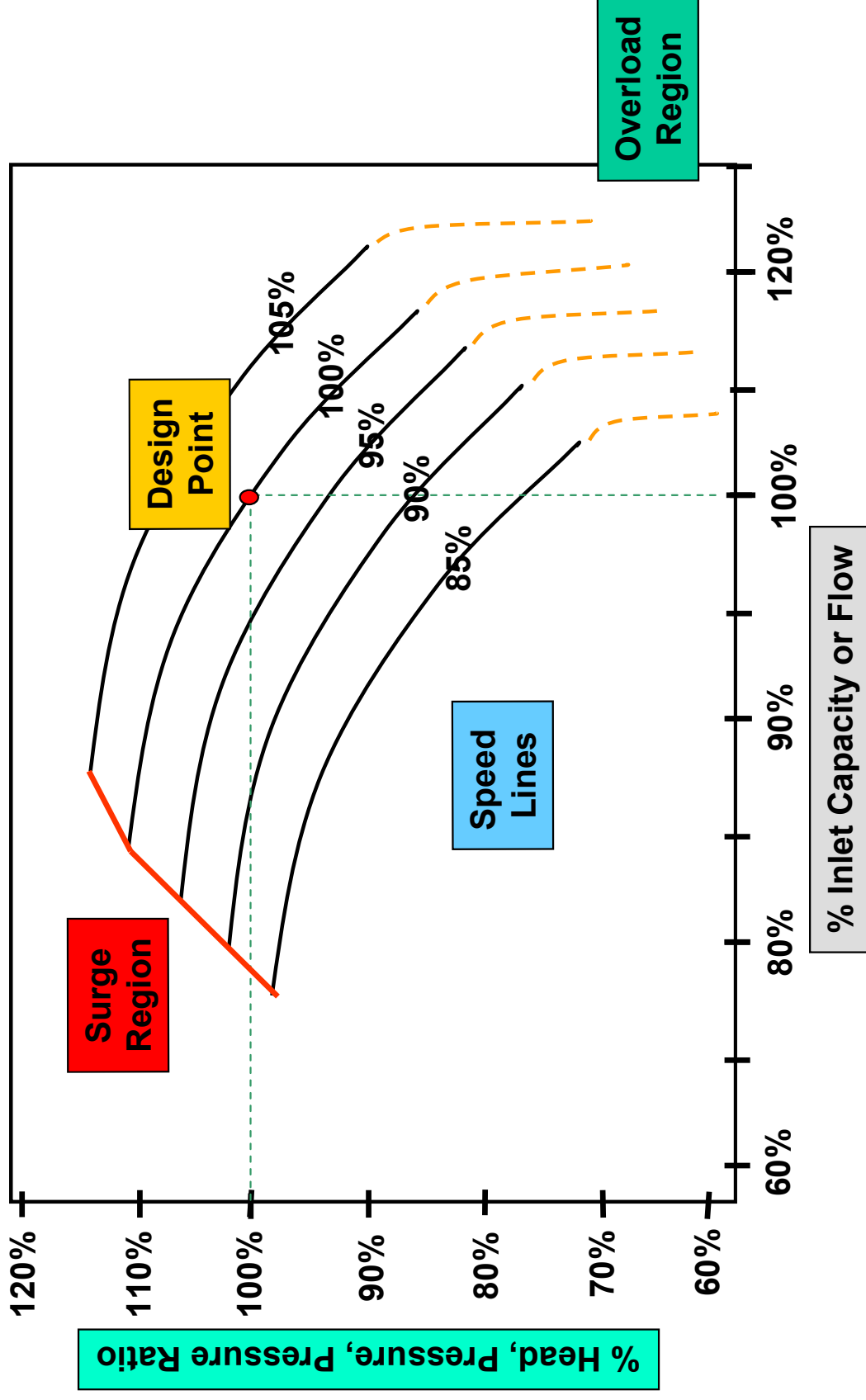
M: Polytrophic Exponent

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Head Concept – Example



Compressor Performance Curves illustrates the operating range and flexibility of a given compressor



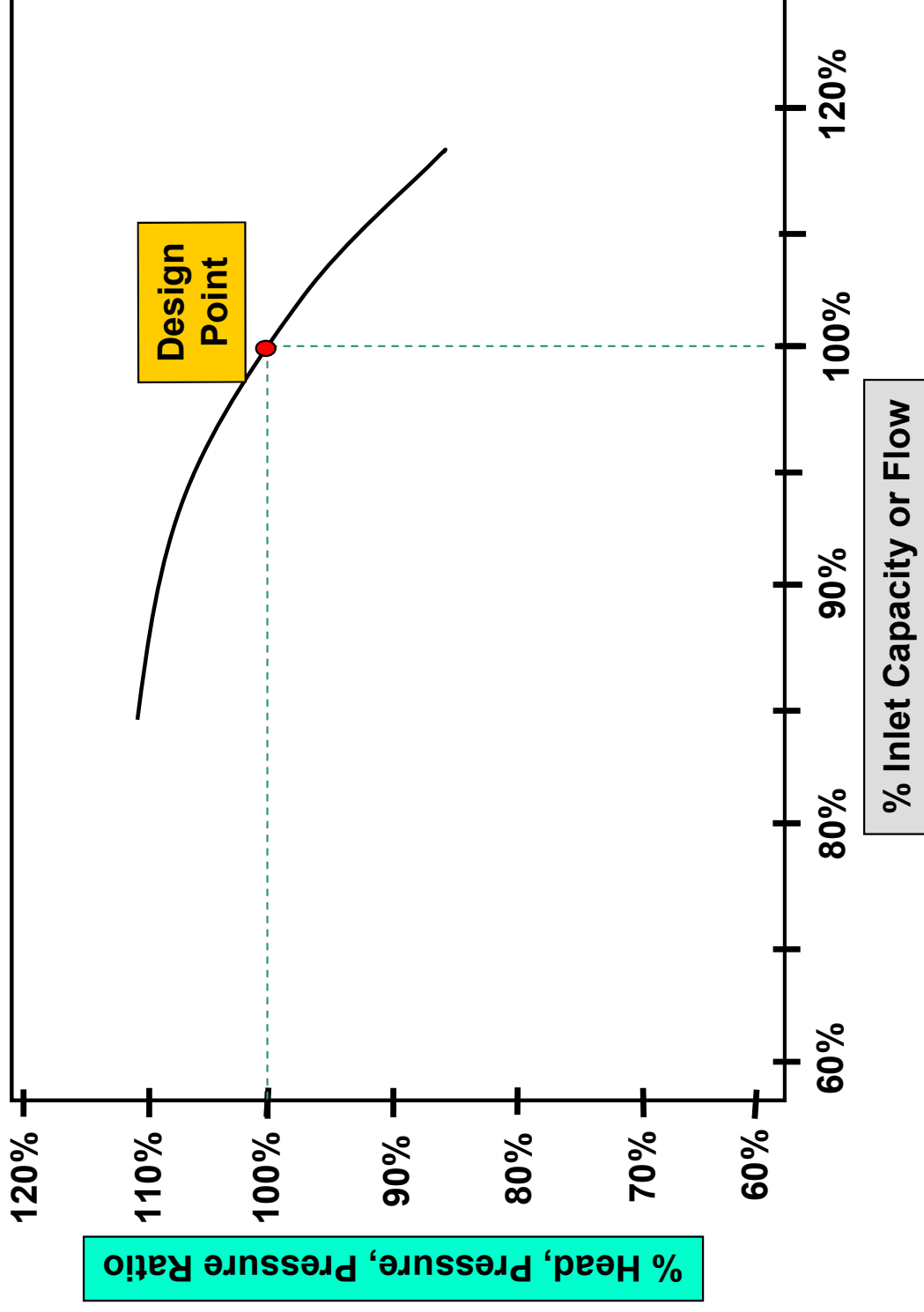
Compressor Performance Curves

- ◆ There are two types of curves that are generally required, section and overall:
 - section refers to an impeller or sequence of impellers between two nozzles such that there is no pressure drop or temp reduction between impellers
 - overall refers to a complete compressor or compressor train

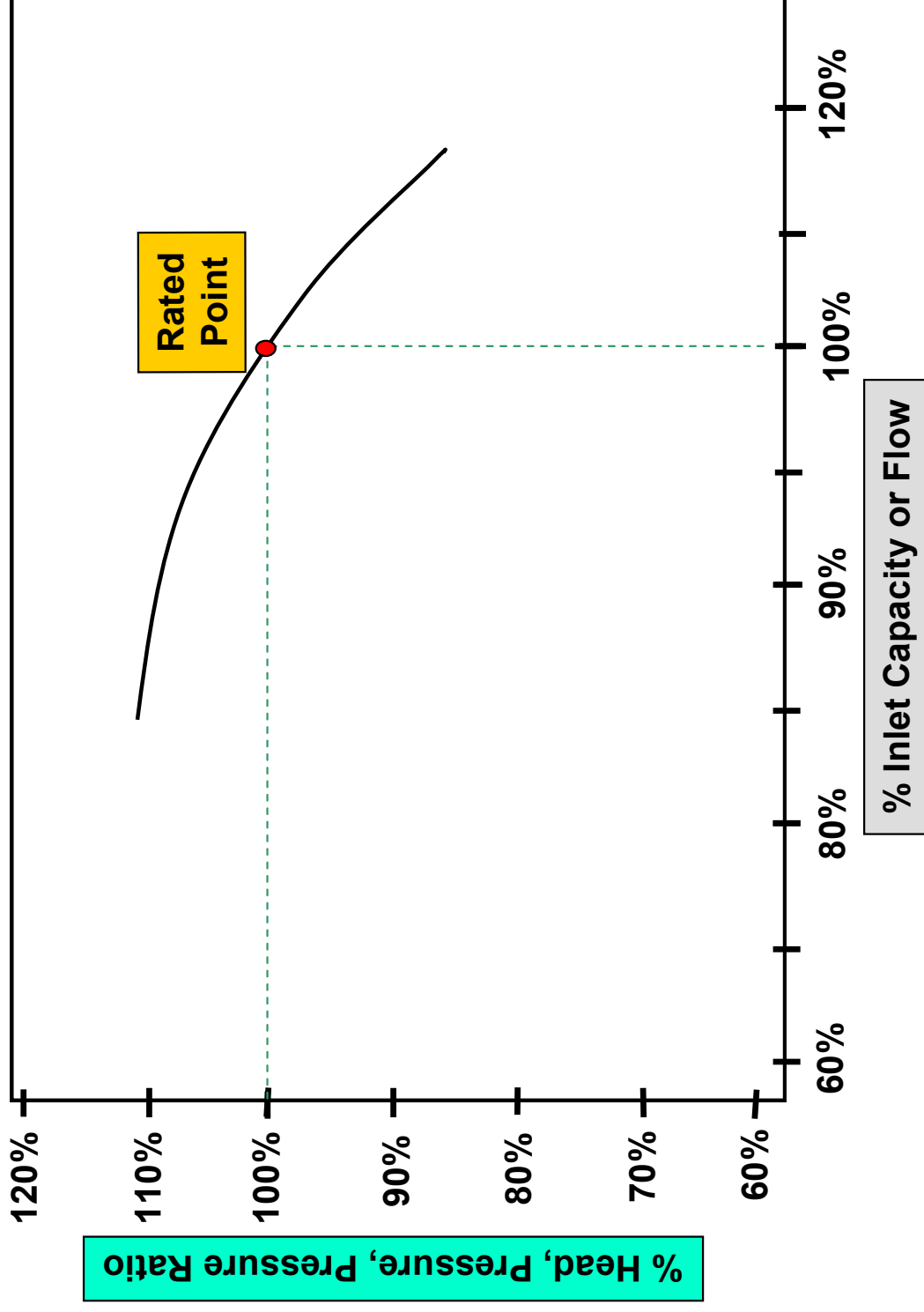
Note: a back-to-back unit with a crossover may often be considered a two-section compressor; but with respect to performance curves, it is a single section since no pressure drop or cooling is introduced between the impellers

- ◆ For single section compressors, the section curves and overall curves are one in the same

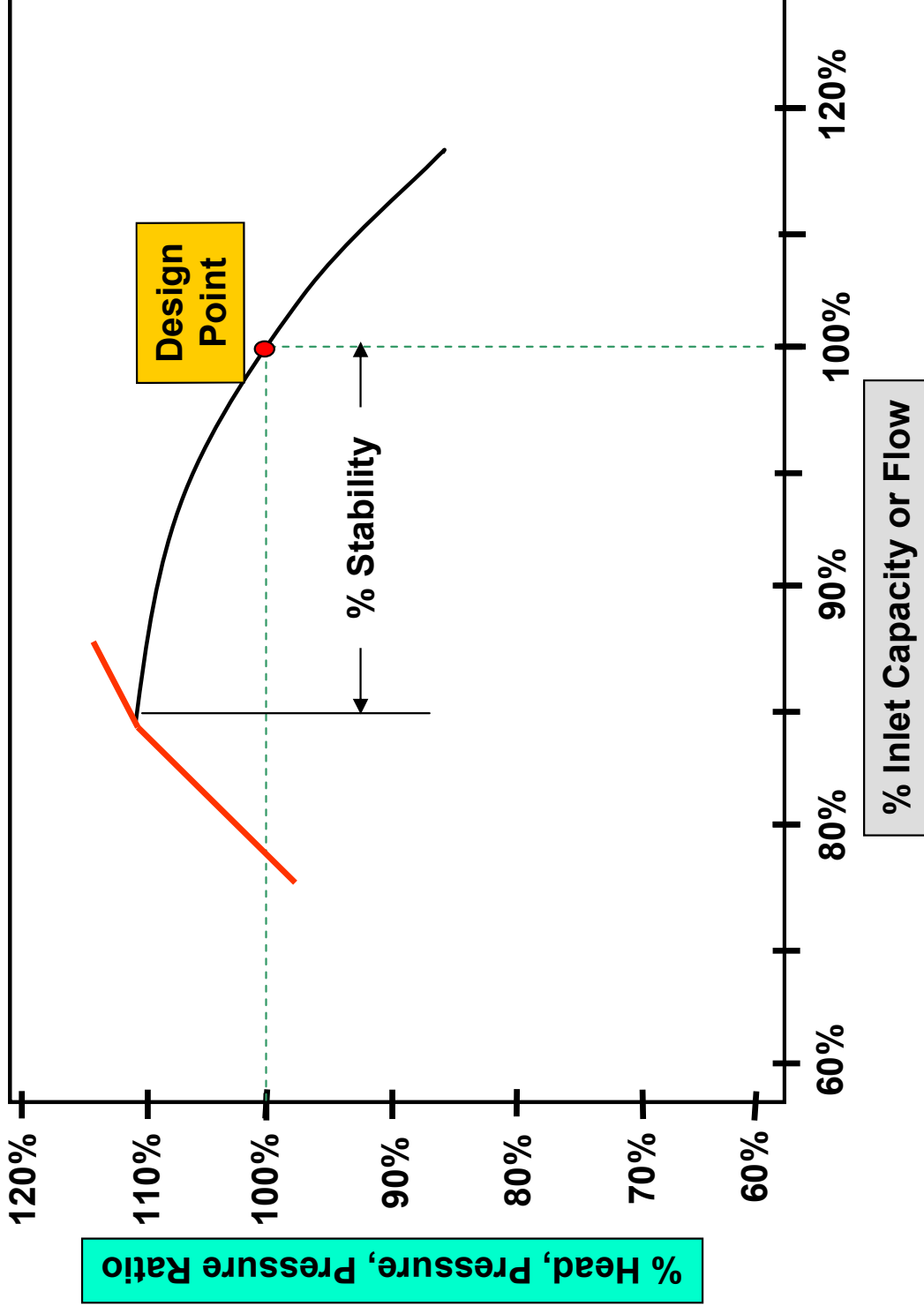
Design Point is the point at which usual operation is expected and optimum efficiency is . It is the point at which the vendor certifies that performance is within the tolerance



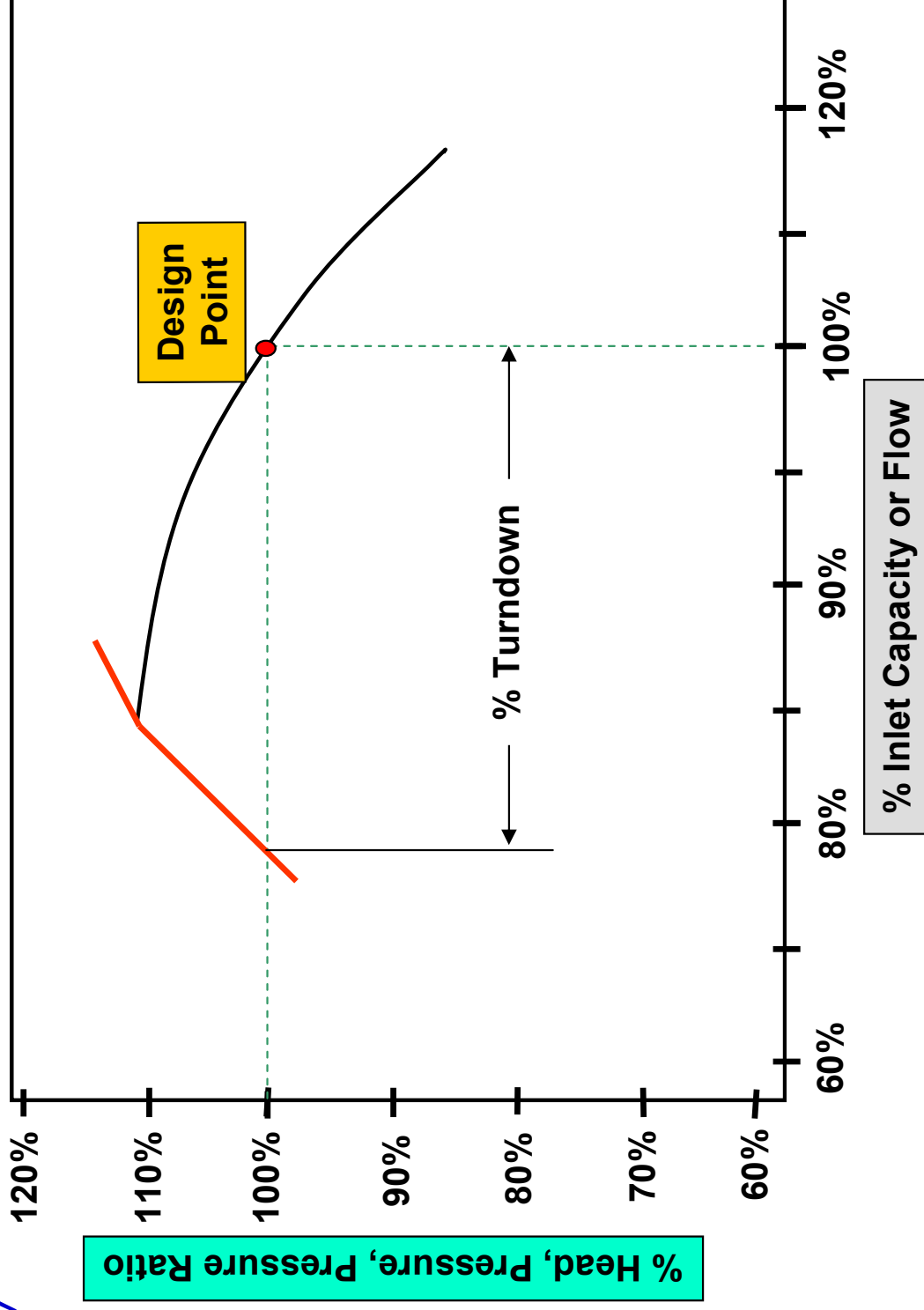
Rated Point is intersection on the 100 % speed line corresponding to the highest flow of any operating point



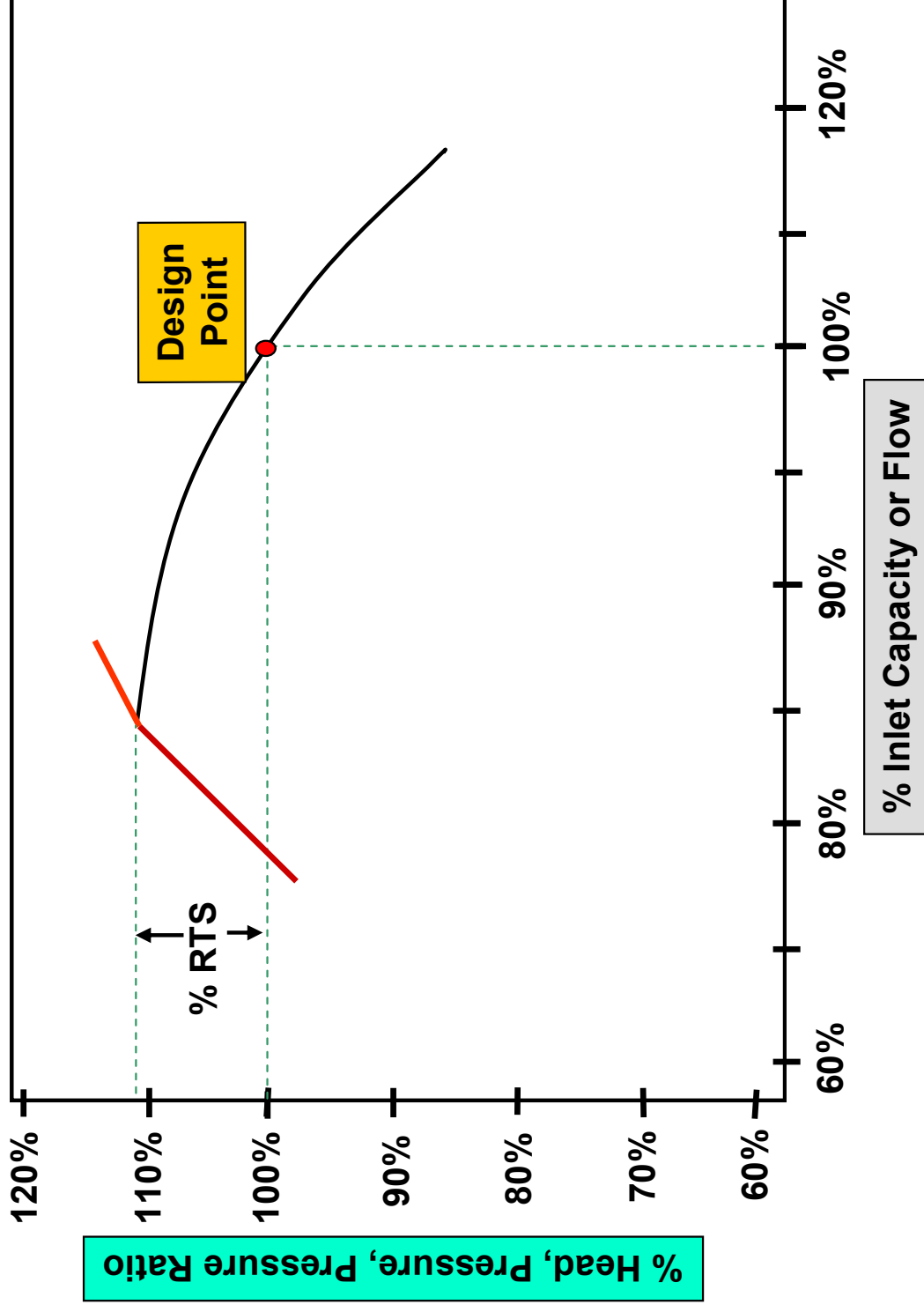
Stability: the percent of change in capacity between the rated (design point) capacity and surge (limit) point, all at constant speed, is measured as the stability of the centrifugal compressor. *Indicates the capability of the centrifugal compressor to operate at less than design flow*



Turndown: the percent of change in capacity between the rated (Design point) capacity and the surge (limit) point, all at constant head or pressure is measured as turndown of the centrifugal compressor. *Indicates the capability of the centrifugal compressor to operate at less than design flow*



Rise to Surge: the percent of change in discharge pressure between the rated point and surge limit at constant speed. High RTS means the compressor can accommodate a modest increase in discharge pressure with a little change in flow



Surge Phenomenon

- ◆ At any given speed, there is minimum flow, below which, the compressor cannot be operated in a stable condition. *This minimum flow value is called “surge” point.*
- ◆ Surge is oscillation of the entire flow of the compressor system and this oscillation can be detrimental to the machine.
- ◆ Compressor surge may be evidenced by the following:
 - a) Excessive rotor vibration
 - b) Increasingly higher process gas temp
 - c) Rapid changes in axial thrust
 - d) Sudden changes in load
 - e) Audible sounds (if surge is severe)

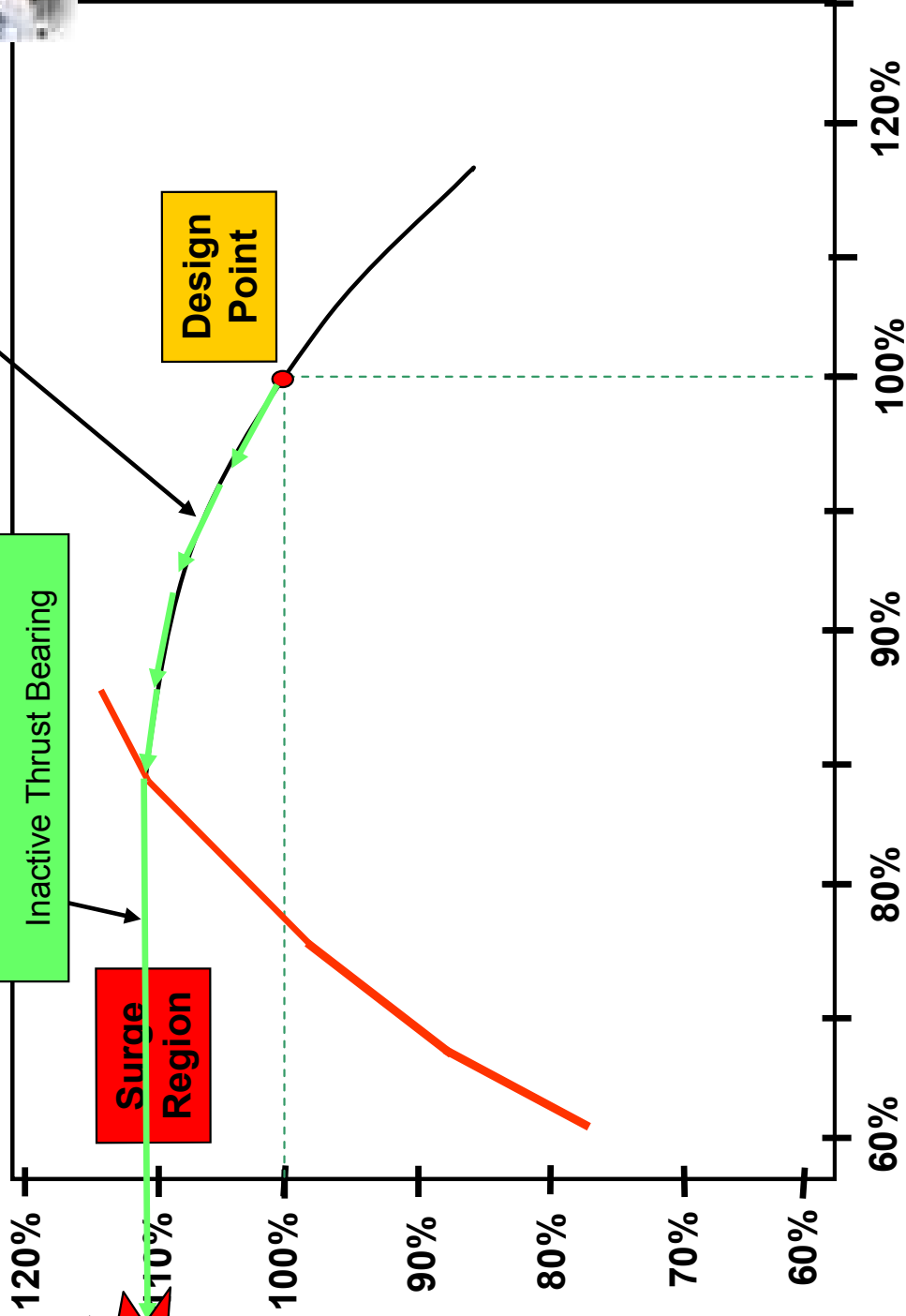
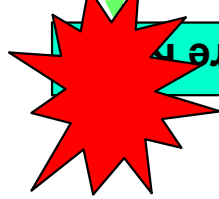
Surge Description

Resistance to Flow Causes Pressure to Rise Which Causes Flow to Decrease

Sudden Reversal of Flow Slams Thrust Disc Against Inactive Thrust Bearing

Surge Region

Design Point

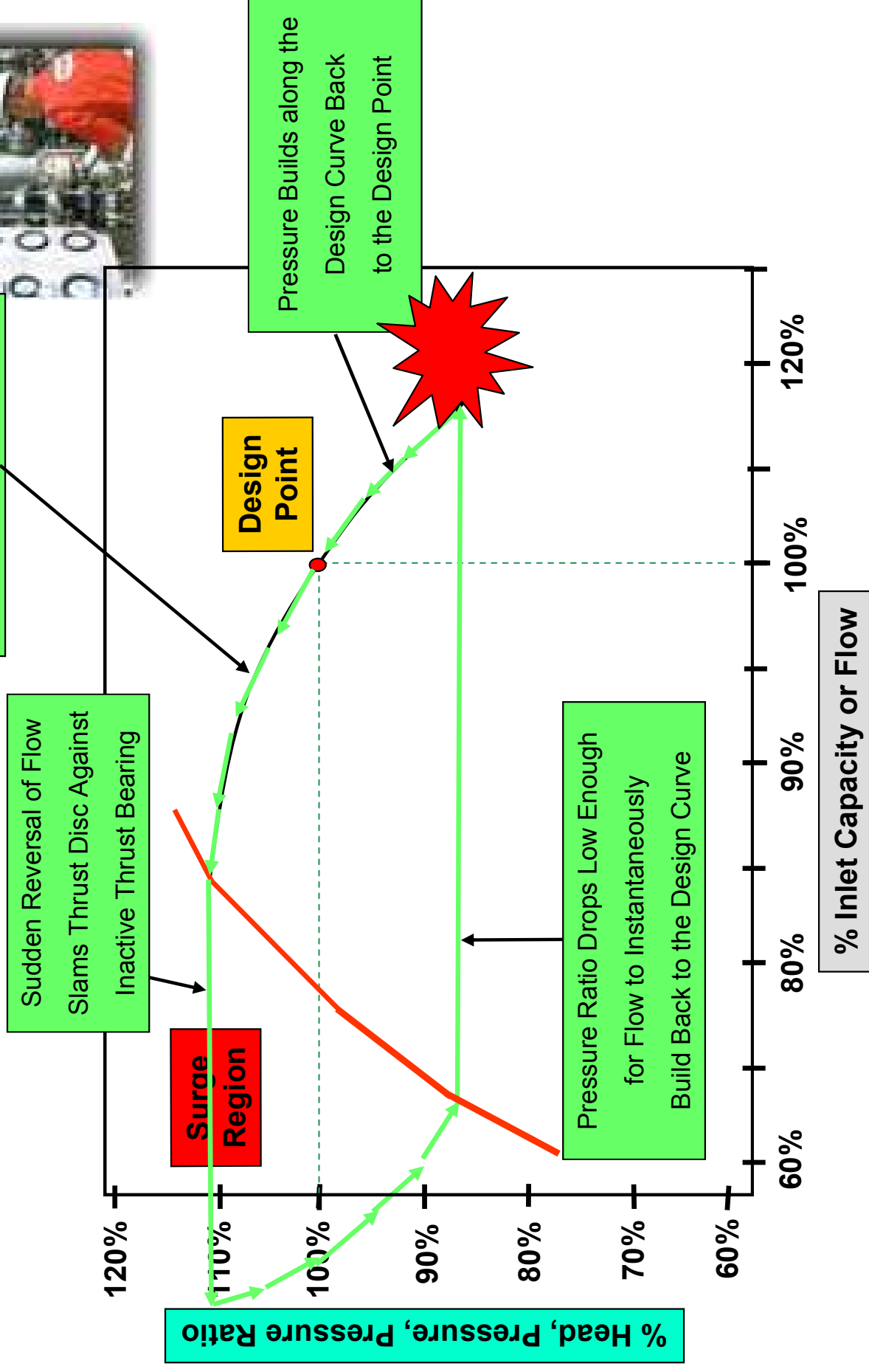
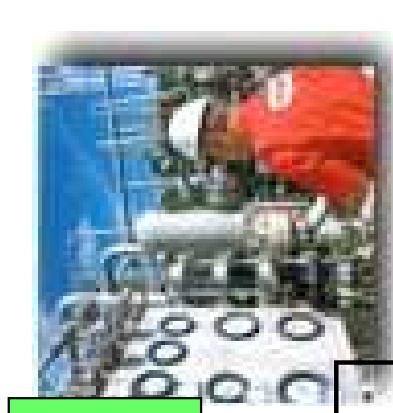


% Inlet Capacity or Flow

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Surge Description



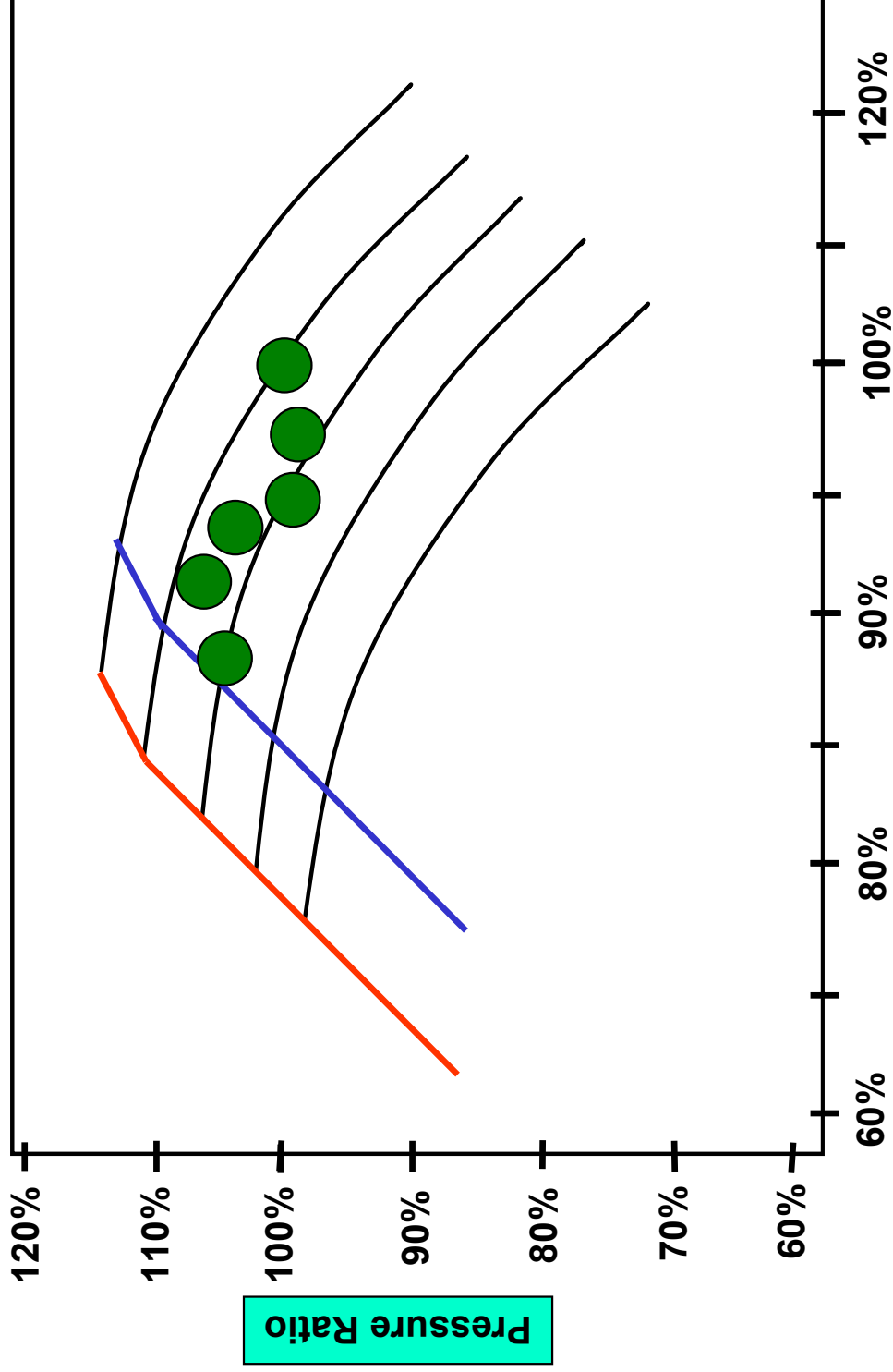


Surge Control on Performance Curves

Operating Point

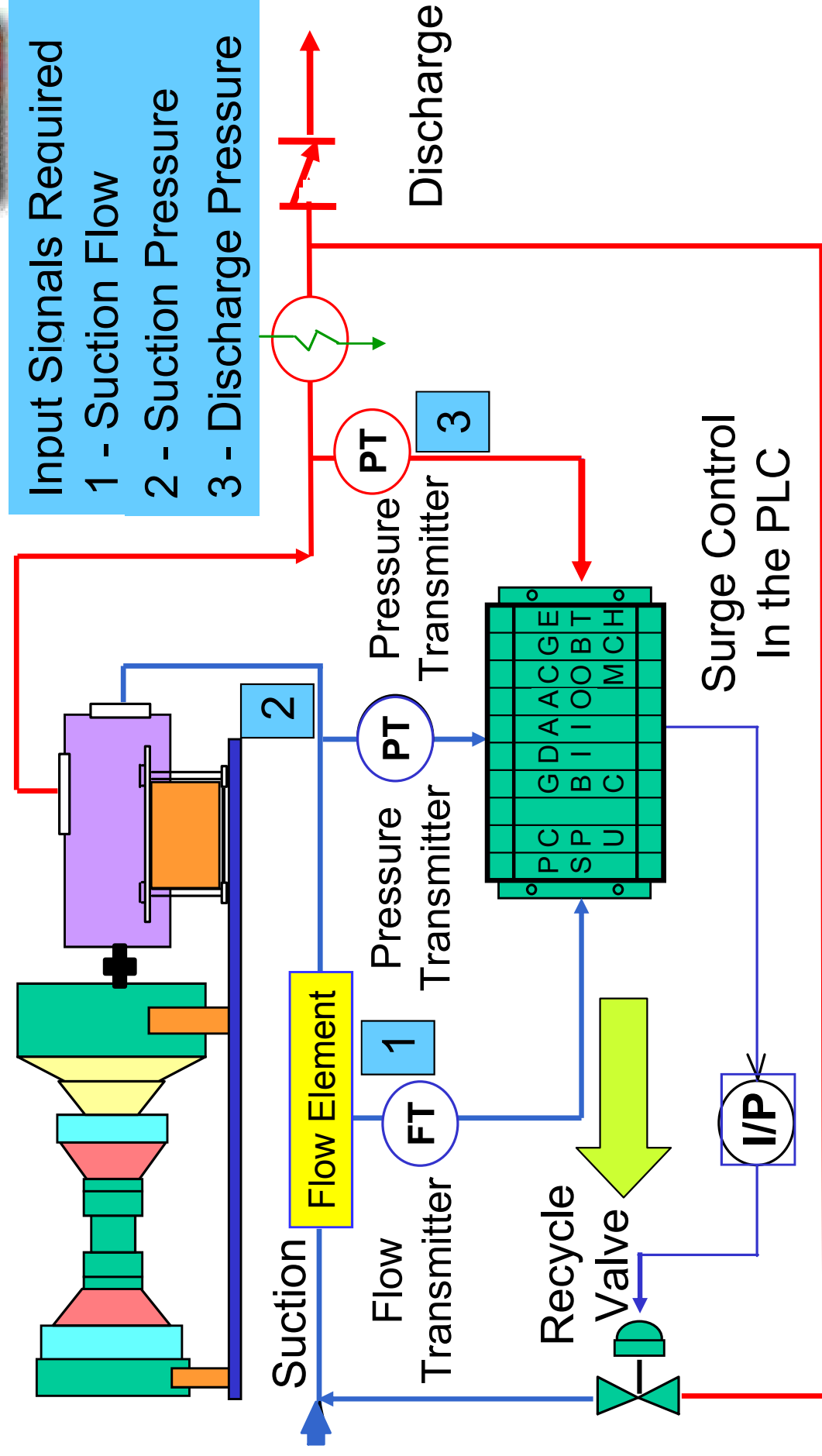
Control Line

Surge Line



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Surge Control System



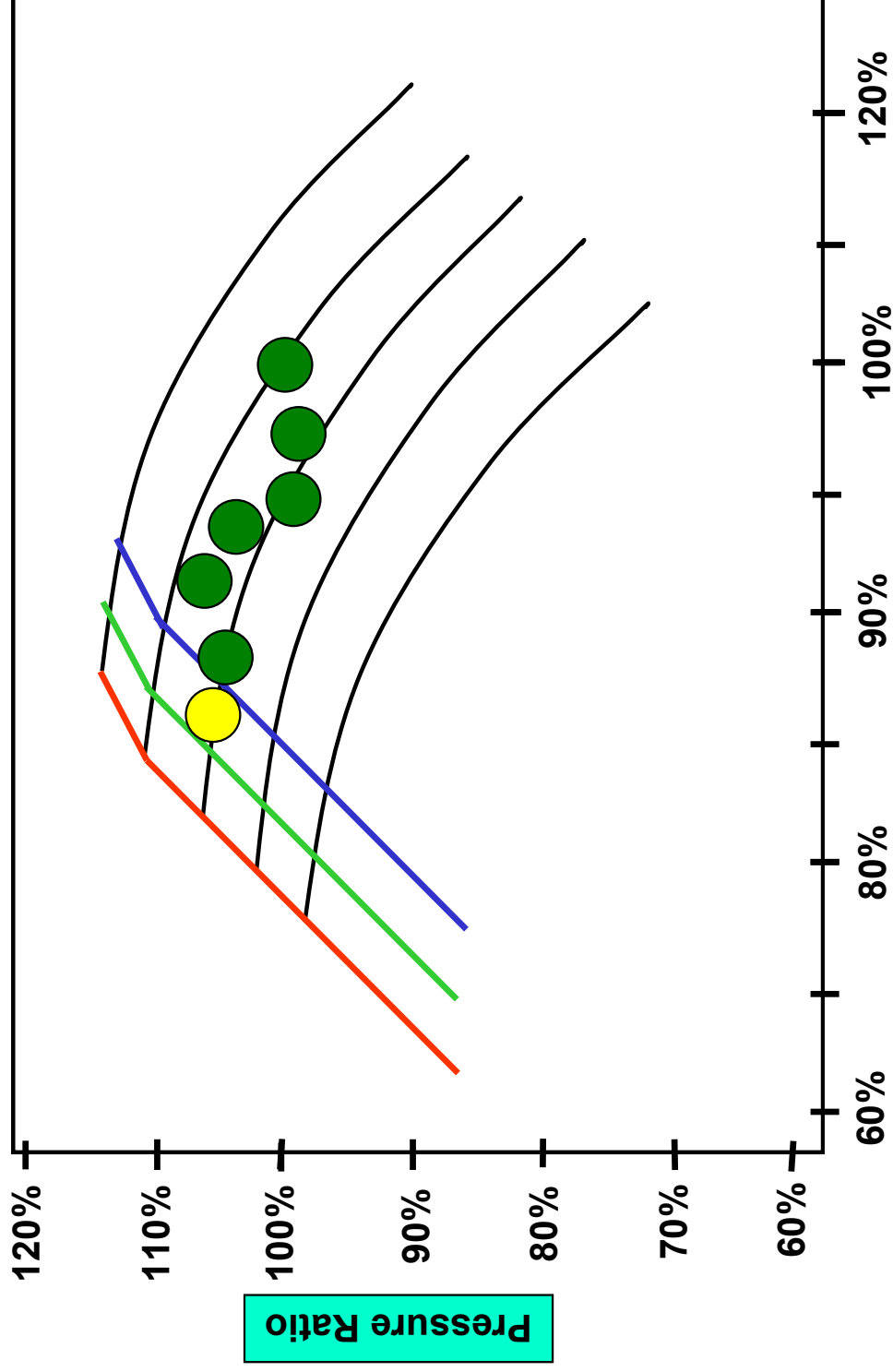
Surge Control on Performance Curves

Operating Point

Control Line

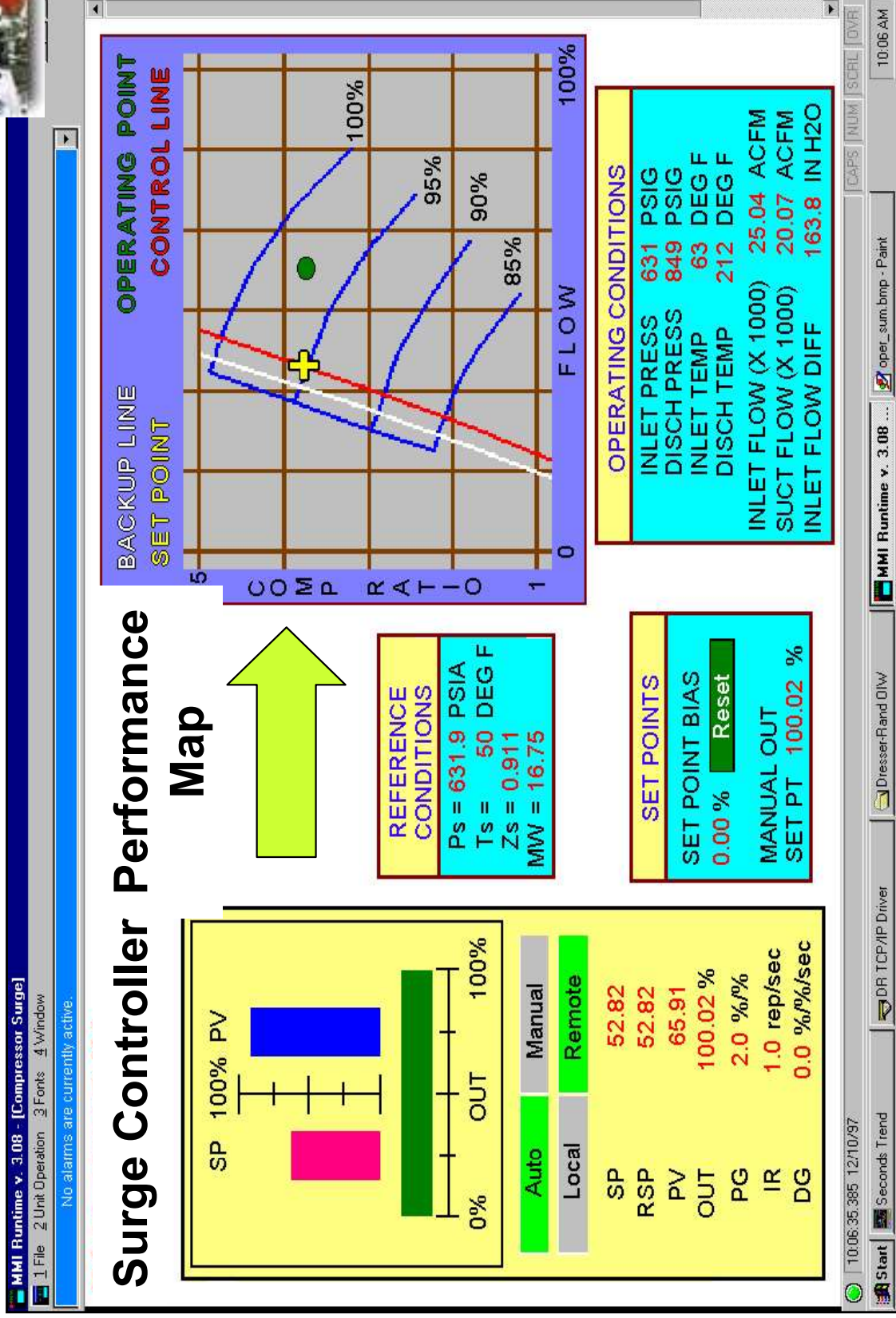
Surge Line

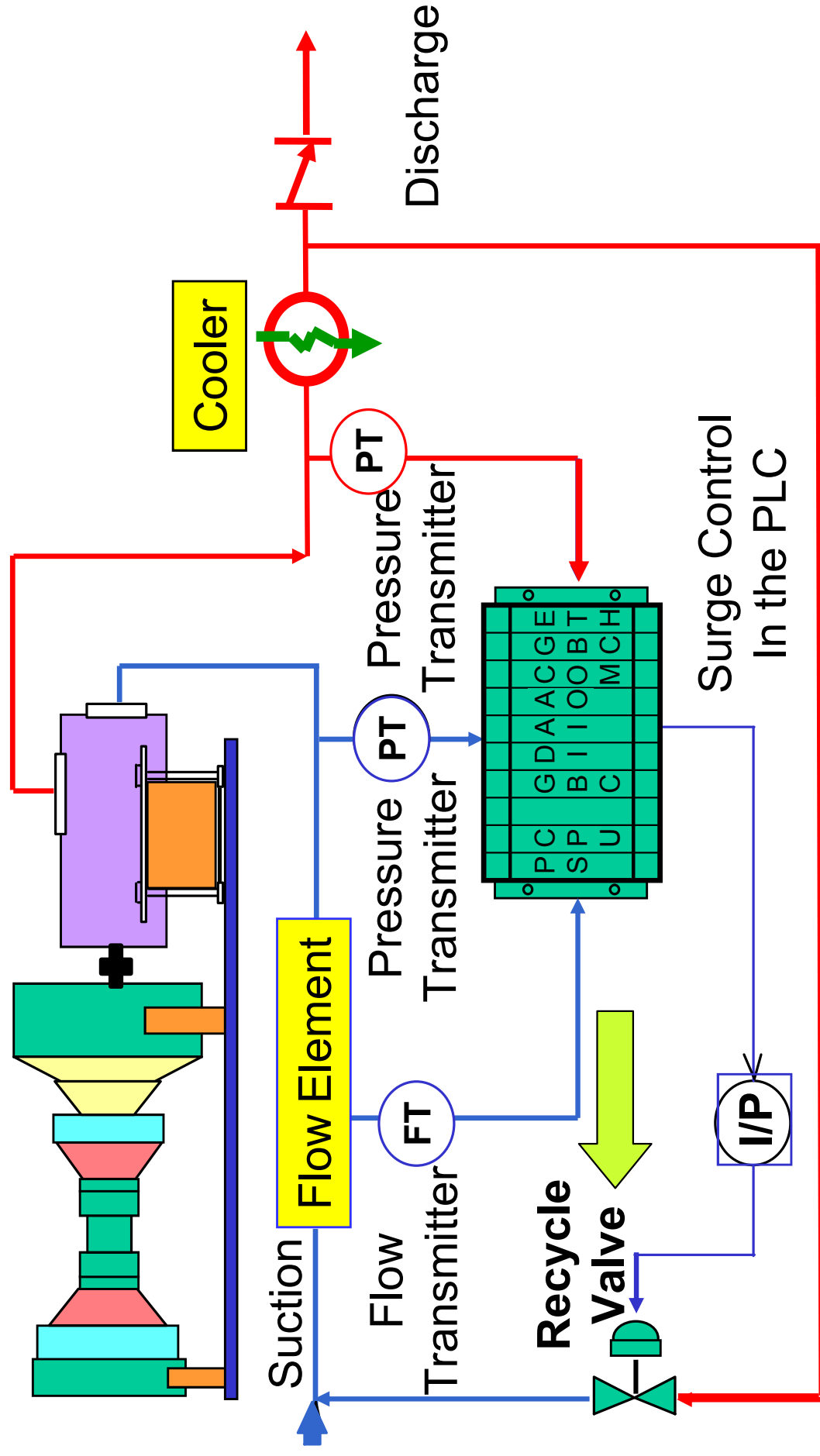
Backup Line



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Surge Control





Surge – Damage of Compressor Internals



Deformation due to high temperature

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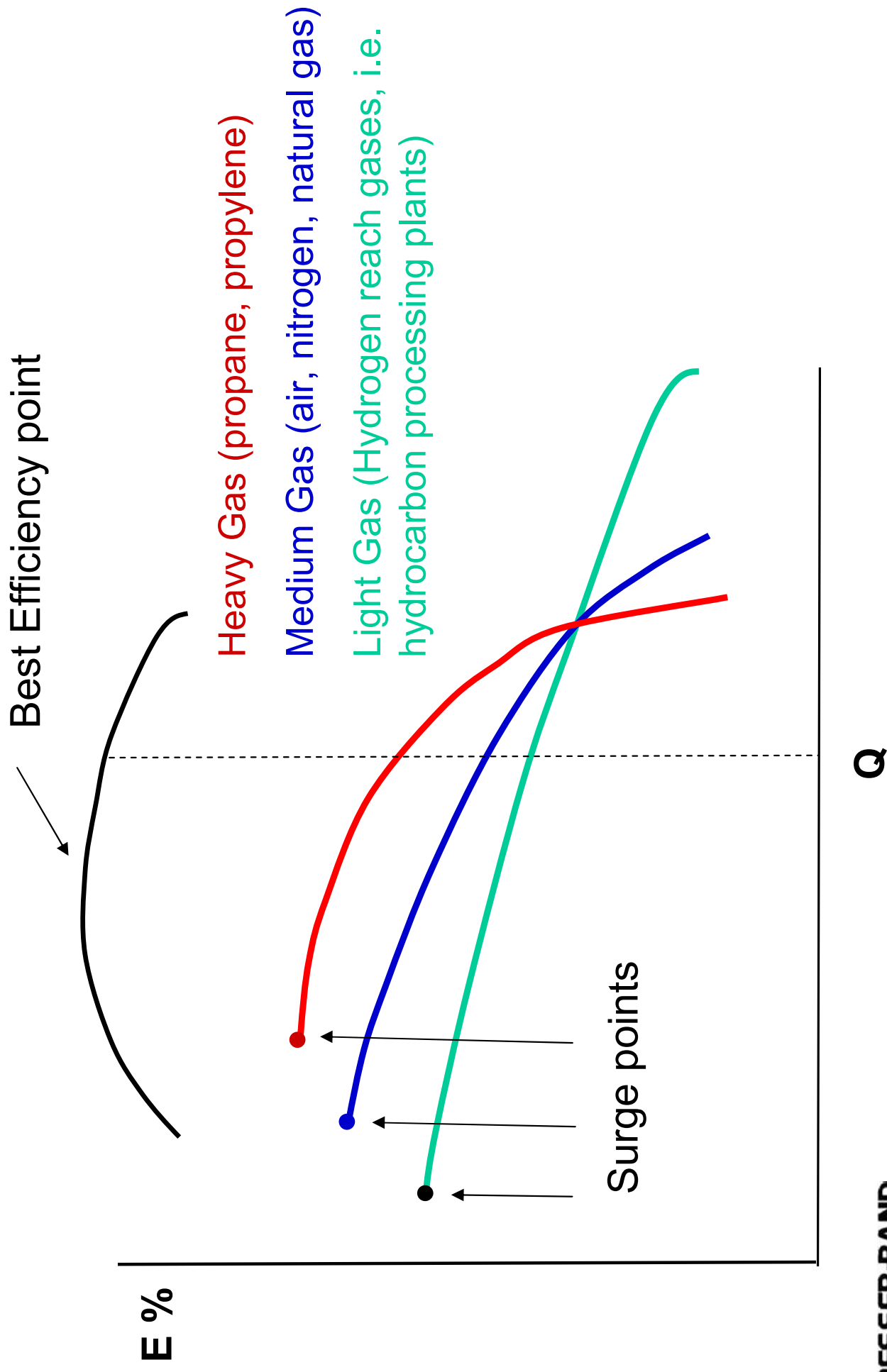
High axial displacement



Surge

- ◆ The frequency of the surge cycle varies inversely with the volume of the system
- ◆ If the check valve is located near compressor discharge nozzle, the frequency will be much higher than of a system with a large volume in the discharge upstream of the check valve
- ◆ The higher frequency of the surge, the intensity will be lower (i.e. few cycles / minute up to more than 20 cycles / sec)
- ◆ *The intensity of the surge increases with gas density , pressure and lower temperature*

Surge - Effects of Gas Composition



Surge - Effects of Gas Composition

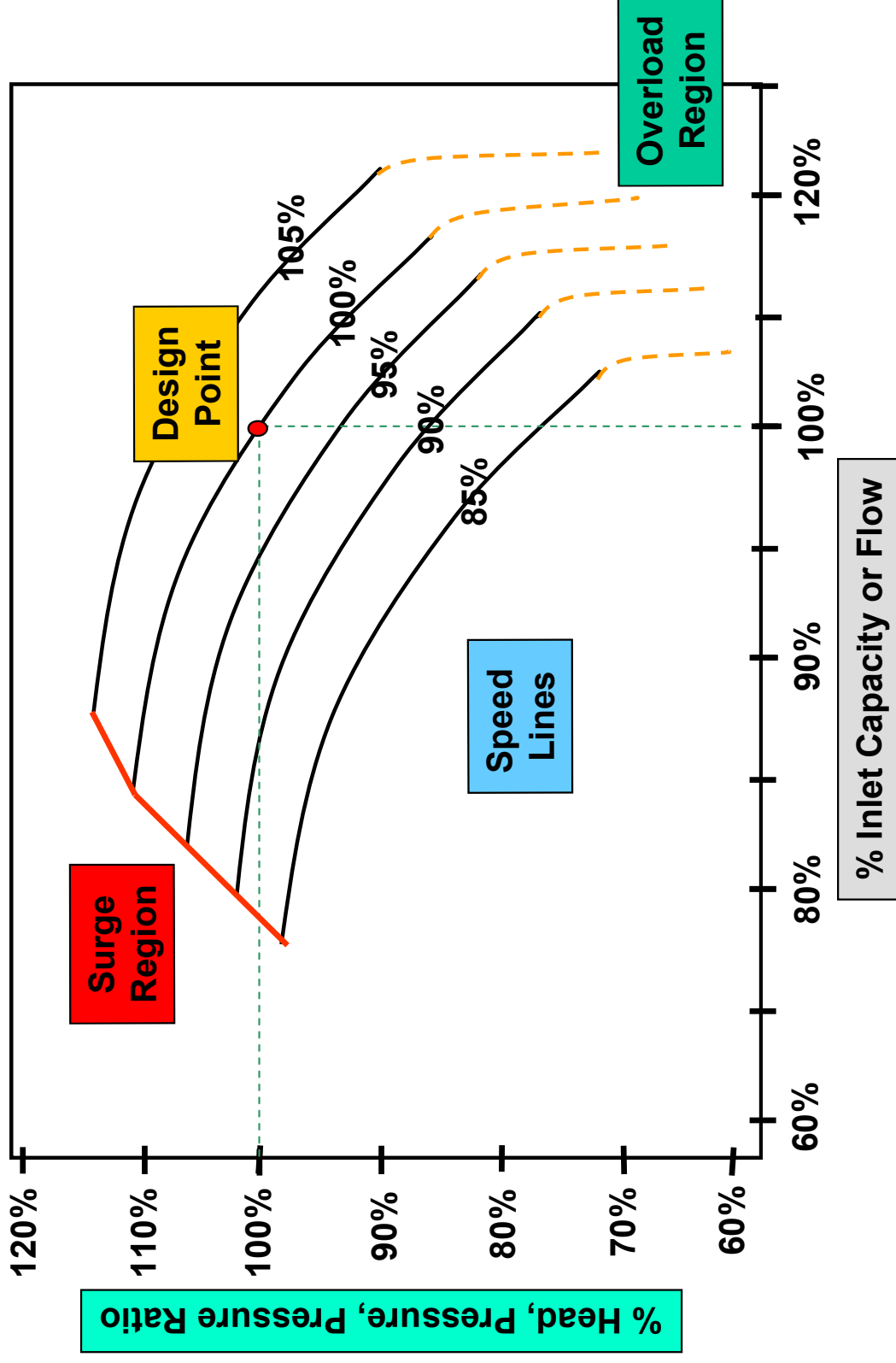
Observations made in respect to the heavy gas:

- ◆ The flow at surge is higher;
- ◆ The stage produces more head than corresponding to medium gas / light gas
- ◆ The right side of the curve turns downward (approaches stonewall) more rapidly
- ◆ The curve is flatter in the opening stage (small RTS)

External Causes and Effects of Surge

- ◆ Restriction in suction or discharge of system
- ◆ Process changes in pressure, temperatures, or gas MW
- ◆ Internal plugging of flow passages of compressor (fouling)
- ◆ Inadvertent loss of speed
- ◆ Instrument or control valve malfunction
- ◆ Operator error
- ◆ Misdistribution of load in parallel operation
- ◆ Improper assembly of compressor (impeller overlap)

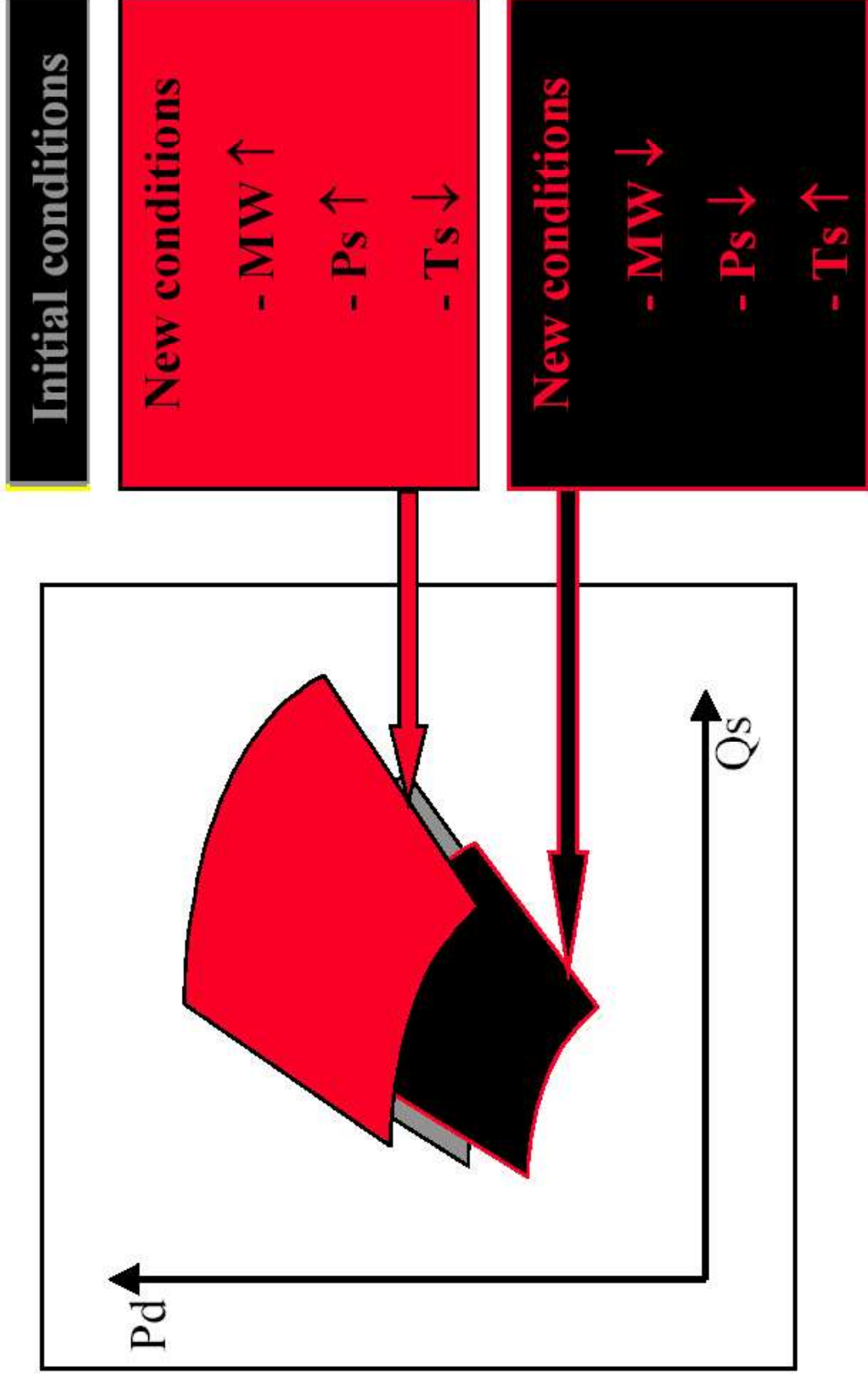
Overload



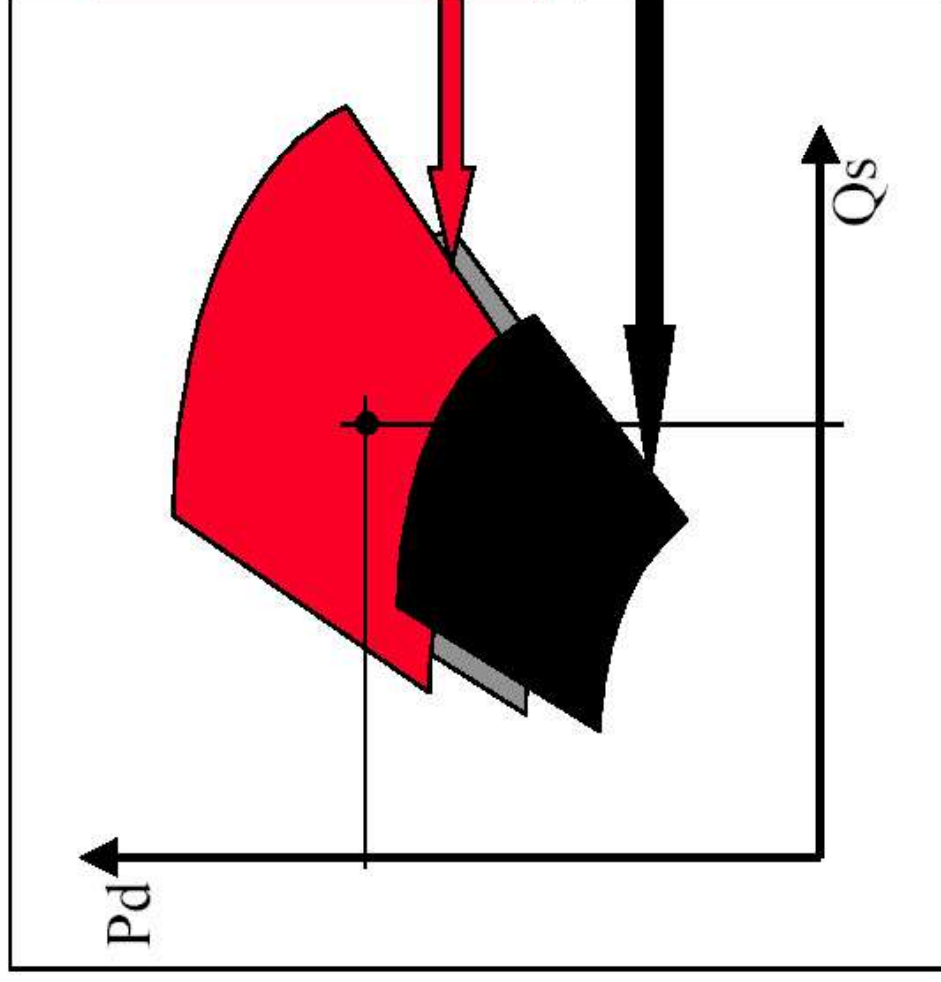
Choke Limit

- ◆ Choke is the maximum flow that a centrifugal compressor can handle at a given speed. At that point, the compressor is unable to produce any net overall pressure ratio.
- ◆ The maximum flow region of the compressor performance curve is where the gas speeds approach *Mach 1*
- ◆ Gas compression is no longer occurring in the compression channels. This region of the curve, as it becomes almost vertical at the choke limit, is also known as “*Stonewall*”
- ◆ Stonewall is usually not detrimental to the compressor, it simply limits the maximum flow. If choke occurs at an off design condition, the maximum volume flow can be increased by increasing the rotational speed

Performance Curves – Inlet Gas Condition Effects



Performance Curves – Inlet Gas Condition Effects



Initial conditions

Case I (MW or $P_s \uparrow$ or $T_s \downarrow$)

Increased operating range,
both for flow and head:

⇒ more flow achievable,

⇒ higher discharge pressure

Case II (MW or $P_s \downarrow$ or $T_s \uparrow$)

Reduced operating range:

⇒ limited by max speed,

⇒ detrimental to operations.

Factors Affecting Compressor Performance

- ◆ MW & Head - If MW increases, the head for a given ratio will decrease in direct proportion
- ◆ Temp & Head - If the Ts increases, the head for a given ratio will increase in direct proportion
- ◆ Zave & Head - If the average compressibility increases, the head will increase in direct proportion
- ◆ N and Head - If speed increases, the head will increase in direct proportion
- ◆ BHP and Head - If Head increases, the BHP will increase in direct proportion
- ◆ Flow and Speed - If the speed increases, the flow will increase in direct proportion

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Factors Affecting Compressor Performance

- ◆ N & BHP - If the speed increases, the BHP will increase in proportion to the cube of the speed. (Because flow increases directly as speed and head increases as the square of the speed and BHP is the product of head X mass flow)
- ◆ Density - The only thing a compressor impeller sees is inlet capacity. Thus to get more capacity out of an existing compressor it is necessary to change the density of the inlet by:
 - decreasing the suction temperature
 - increasing the suction pressure
 - increasing the MW of the gas

Compressor Off - Design Performance

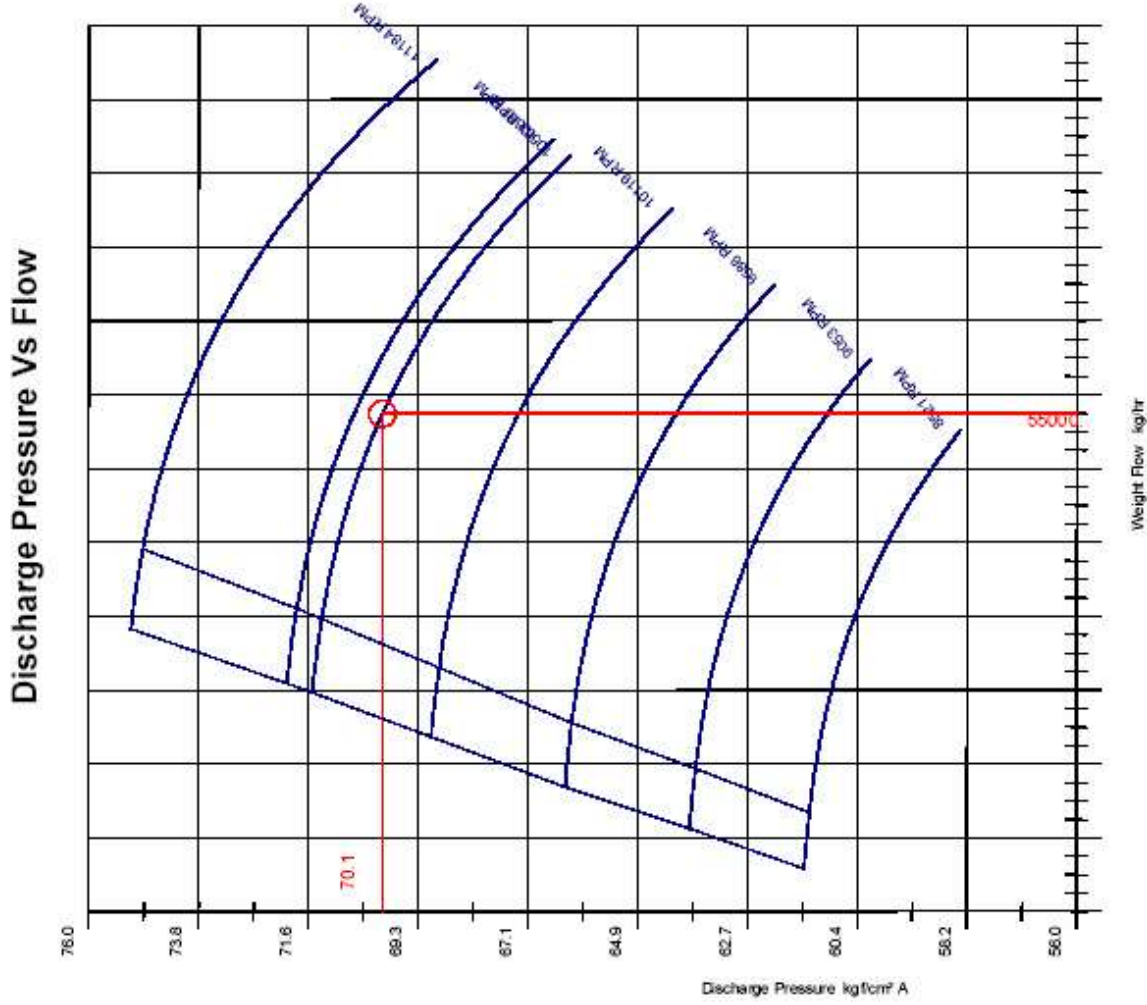
- ◆ Performance curves for axial and centrifugal compressors are usually based on **constant inlet conditions (P_s , T_s , MW)**. In actual service, these compressors rarely see these base curve conditions exactly
- ◆ If the field inlet conditions deviate more than 5% from the curve inlet conditions then the field data can not be accurately plotted on the curve without converting the field data to curve conditions
- ◆ To properly evaluate the compressor (running off design), the performance parameters shall be corrected to the design conditions

Corrected Mass and Volume Flow

$$\frac{Q_{act} * \sqrt{\frac{MW_{act}}{MW_{ref}}}}{\sqrt{\frac{T_1}{T_{ref}}}} = \frac{M_{act} * \sqrt{\frac{T_1}{T_{ref}}}}{\frac{P_1}{P_{ref}} \sqrt{\frac{MW_{act}}{MW_{ref}}}}$$

- ◆ M_{act} Q_{act} - actual mass & volumetric flow;
- ◆ P_{ref} , T_{ref} , MW_{ref} – Inlet pressure, Inlet Temperature, Gas MW, at reference conditions (design inlet conditions);
- ◆ P_1 , T_1 – Inlet Pressure, Inlet Temperature at actual operating conditions;

Operation Limitations



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Compressor

Driver

Power

Process

Real-Time Data Historical Data

ASG

Gas Mixture: ASG

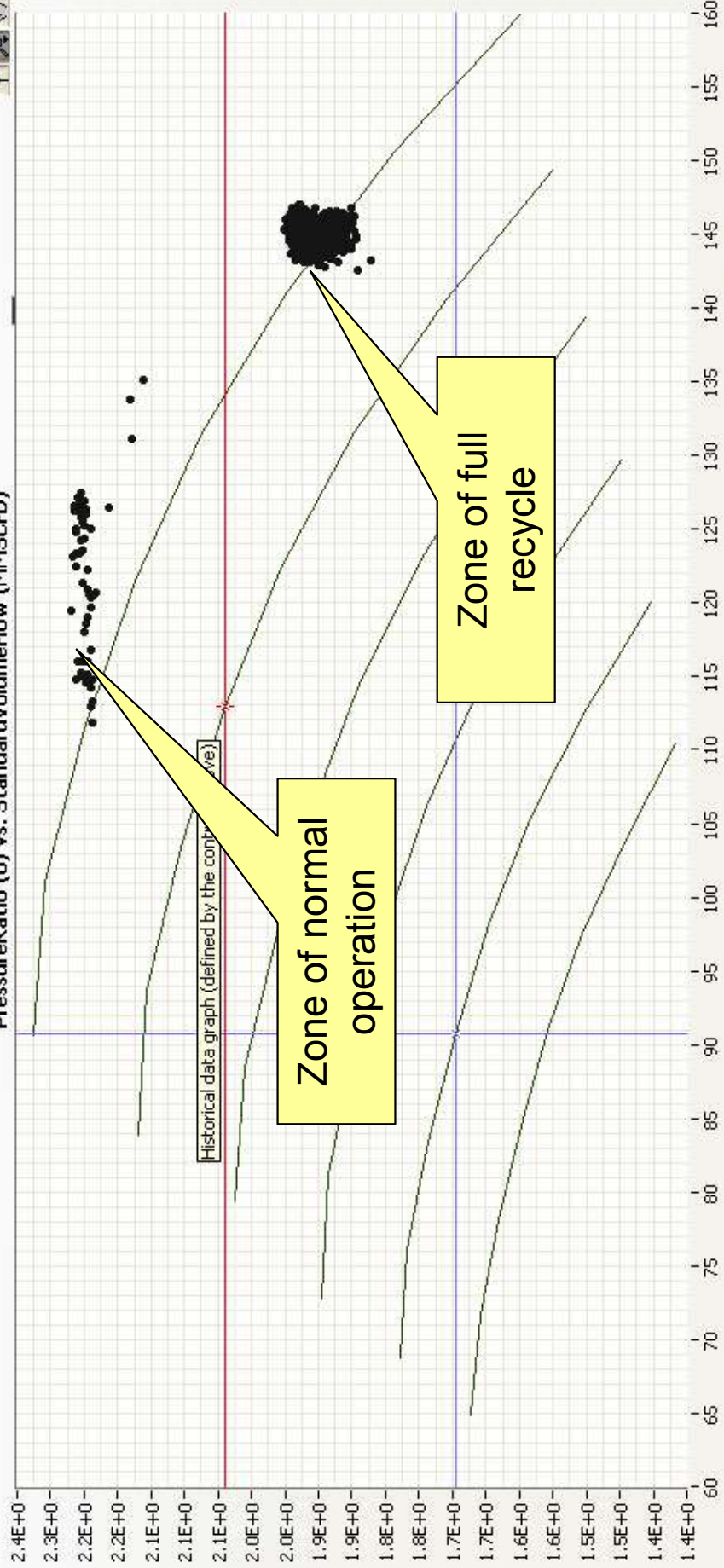
Graph: PR vs. Standard Volume

From: 04:20 PM 09/01/06

To: 07:11 AM 09/03/06

Update

PressureRatio (0) vs. StandardVolumeFlow (MMSCFD)



Cursor 0 112.89 2.04 Cursor 1 90.776 1.70

Maximum: 2.27 0 Mean: 1.93 0 Minimum: 1.82 0

Print

Save

Exit

Real-Time Data Historical Data

ASG

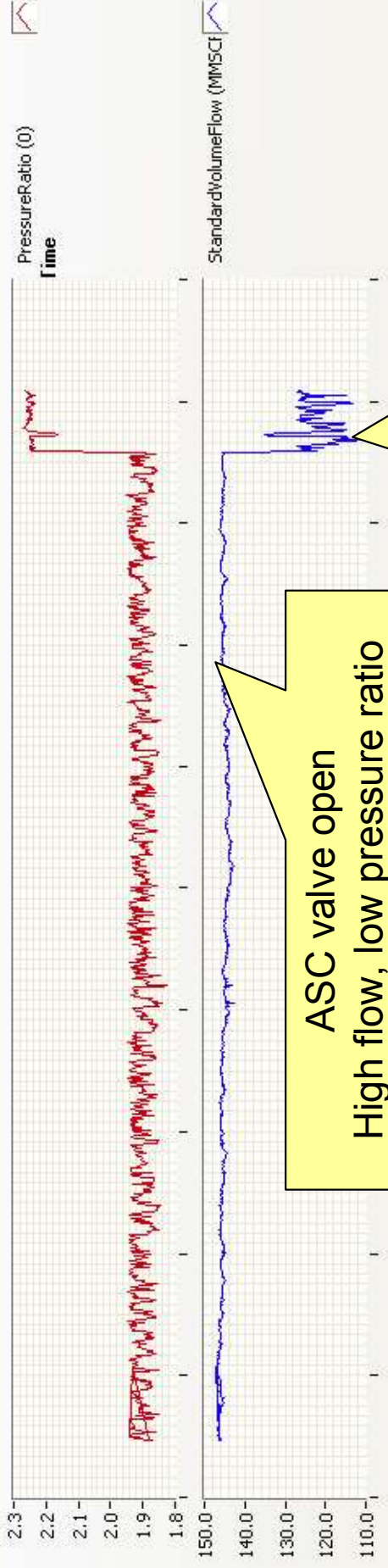
Gas Mixture: ASG

Graph: Trend Select Traces...

From: 04:20 PM 09/01/06

To: 07:11 AM 09/03/06

Update



Historical data graph (defined by the controls above)

Time	PressureRatio (0)	StandardVolumeFlow (MMSCF)
1:00:00 PM 09/02/06	1.93	120
3:00:00 PM 09/02/06	1.93	120
5:00:00 PM 09/02/06	1.93	120
7:00:00 PM 09/02/06	1.93	120
9:00:00 PM 09/02/06	1.93	120
11:00:00 PM 09/02/06	1.93	120
1:00:00 AM 09/03/06	1.93	120
3:00:00 AM 09/03/06	1.93	120
5:00:00 AM 09/03/06	1.93	120
7:00:00 AM 09/03/06	1.93	120
9:00:00 AM 09/03/06	1.93	120

Maximum: 2.27 0

Mean: 1.93 0

Minimum: 1.82 0

Save

Print

Exit

Compressor Operation Issues- Efficiency Drop

- ◆ Internal recycle
- ◆ Un-tuned Surge Control System
- ◆ Leakage via by-pass valve(s) in process
- ◆ Compressor operated out of “*guaranteed performance envelope*”
- ◆ Impeller & Diaphragm erosion
- ◆ Fouling